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Loading and unloading by an autonomous mobile robot

Abstract

To load and unload a trailer, an autonomous mobile robot determines its location and the location of objects within the trailer relative to the trailer itself, rather than relative to a warehouse. The autonomous mobile robot determines its location the location of objects within the trailer relative to the trailer. The autonomous mobile robot navigates within the trailer and manipulates objects within the trailer from the trailer's reference frame. Additionally, the autonomous mobile robot uses a centerline heuristic to compute a path for itself within the trailer. A centerline heuristic evaluates nodes within the trailer based on how far away those nodes are from the centerline. If the nodes are further away from the centerline, they are assigned a higher cost. Thus, when the autonomous mobile robot computes a path, the path is more likely to stay near the centerline of the trailer rather than get closer to the sides.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/295,023, filed Dec. 30, 2021, which is incorporated by reference.

BACKGROUND
(1) An autonomous mobile robot is a robot that is capable of autonomously navigating an environment (e.g., a warehouse environment) and manipulating objects within that environment. An autonomous mobile robot may use a 2D environment map to navigate within a warehouse environment. The environment map may include information about objects and obstacles within the warehouse. For example, the environment map may contain information about the poses of objects within the warehouse, and the autonomous mobile robot may use that information when manipulating the objects within the warehouse.
(2) However, an environment map may not be sufficient when the autonomous mobile robot enters a new environment, like a trailer. For example, a trailer may have some roll or pitch with respect to the warehouse floor because a trailer bed may not be perfectly parallel to the ground and/or because a trailer may not be backed flush against a loading bay. These inaccuracies in the position of the trailer mean that the trailer may not be properly integrated into the environment map, which could cause autonomous mobile robots using the environment map within the trailer to inaccurately navigate within the trailer. Because the trailer is an environment with little room to maneuver, these inaccuracies may render the autonomous mobile robot unable to effectively load or unload the trailer.
(3) Additionally, an autonomous mobile robot may use a pathfinding algorithm to navigate within

an environment. Traditional pathfinding algorithms typically prioritize identifying a shortest or fastest path from an initial position to a new position. However, the shortest or fastest path may not be ideal for a robot operating within a trailer. For example, if the autonomous mobile robot is traveling to a spot near a side of the trailer, traditional pathfinding algorithms may generate a path that brings the autonomous mobile robot close to the sides of the trailer for a portion of the path. These paths can be problematic because the autonomous mobile robot then has less space to maneuver to manipulate an object or avoid an obstacle. Thus, traditional pathfinding algorithms fail to suffice for autonomous mobile robots operating in crowded or small environments such as trailers.

SUMMARY

(4) To load and unload a trailer, an autonomous mobile robot determines its location and the location of objects within the trailer relative to the trailer itself, rather than relative to a warehouse. The autonomous mobile robot collects sensor data of the trailer when the robot reaches a docking point to which the trailer has docked. The autonomous mobile robot may determine characteristics of the trailer based on the sensor data, such as the width, height, depth, centerline, off-center parking, yaw, roll, or pitch of the trailer. The autonomous mobile robot determines its location relative to the trailer. For example, the autonomous mobile robot may determine its location relative to a centerpoint of the trailer or relative to some part of the trailer. The autonomous mobile robot may also determine the location of objects within the trailer relative to the trailer. The autonomous mobile robot then uses its location and the location of objects within the trailer to navigate within the trailer and manipulate objects within the trailer from the trailer's reference frame.

(5) Additionally, the autonomous mobile robot uses a centerline heuristic to compute a path for itself within the trailer. The autonomous mobile robot uses sensor data to determine an initial pose for itself as well as an end pose for the robot within the trailer. The autonomous mobile robot computes a centerline for the trailer based on the sensor data and computes a path for itself from the initial pose to the end pose. The robot computes the path using a centerline heuristic, which is a heuristic that evaluates nodes within the trailer based on how far away those nodes are from the centerline. If the nodes are further away from the centerline, they are assigned a higher cost value than if they are closer to the centerline. Thus, when the autonomous mobile robot computes a path, the path is more likely to stay near the centerline of the trailer rather than get closer to the sides. Thus, when the autonomous mobile robot travels along the path to the end pose, the autonomous mobile robot generally has more clearance around itself, which improves its ability to navigate and manipulate objects within the trailer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates an environment for operating an autonomous mobile robot using a central communication system, in accordance with some embodiments.

(2) FIG. **2** illustrates exemplary modules and data stores used by a central communication system, in accordance with some embodiments.

(3) FIG. **3** illustrates exemplary modules and data stores used by the autonomous mobile robot, in accordance with some embodiments.

(4) FIG. **4** illustrates an example autonomous mobile robot navigating through a warehouse environment to a trailer, in accordance with some embodiments.

(5) FIG. **5**A illustrates example characteristics that the autonomous mobile robot may determine based on sensor data, in accordance with some embodiments.

(6) FIG. **5**B illustrates example locations of the autonomous mobile robot and objects within the

trailer, as determined by the autonomous mobile robot, in accordance with some embodiments.
(7) FIG. **6**A illustrates an autonomous mobile robot approaching a ramp that leads from the warehouse to the trailer, in accordance with some embodiments.
(8) FIG. **6**B illustrates an autonomous mobile robot attempting to unload an object in the last row of the trailer without adjusting its forklift, in accordance with some embodiments.
(9) FIG. **6**C illustrates an autonomous mobile robot that adjusts its forklift to be level with the floor of the trailer, in accordance with some embodiments.
(10) FIG. **7** illustrates an example autonomous mobile robot that uses a centerline heuristic to navigate within a trailer, in accordance with some embodiments.
(11) FIG. **8** is a flowchart for an example method for automated trailer loading and unloading, in accordance with some embodiments.
(12) FIG. **9** is a flowchart for an example method for pathfinding using a centerline heuristic, in accordance with some embodiments.
DETAILED DESCRIPTION
(13) System Environment
(14) FIG. **1** illustrates an environment for operating an autonomous mobile robot using a central communication system, in accordance with some embodiments. Environment **100** includes operator device **110**, network **120**, central communication system **130**, and autonomous mobile robot **140**. Environment **100** may be generally described herein as a warehouse environment, however, environment **100** may be any kind of environment. Environment **100** need not be limited to a defined space (e.g., an interior of a warehouse), and may include any areas that are within the purview of instructions of an autonomous mobile robot (e.g., parking lots, loading docks, and so on that are outside of a warehouse space). While operator device **110** and central communication system **130** are depicted as being within environment **100**, this is merely for convenience; these devices may be located outside of environment **100** (e.g., at a home, office, data center, cloud environment, etc.). Additional embodiments may include more, fewer, or different components from those illustrated in FIG. **1**, and the functionality of each component may be divided between the components differently from the description below. Additionally, each component may perform their respective functionalities in response to a request from a human, or automatically without human intervention.
(15) Operator device **110** may be any client device that interfaces one or more human operators with one or more autonomous mobile robots of environment **100** and/or central communication system **130**. Exemplary client devices include smartphones, tablets, personal computers, kiosks, and so on. While only one operator device **110** is depicted, this is merely for convenience, and a human operator may use any number of operator devices to interface with autonomous mobile robots **140** or the central communication system **130**. Operator device **110** may have a dedicated application installed thereon (e.g., downloaded from central communication system **130**) for interfacing with the autonomous mobile robot **140** or the central communication system **130**. Alternatively, or additionally, operator device **110** may access such an application by way of a browser. References to operator device **110** in the singular are done for convenience only, and equally apply to a plurality of operator devices.
(16) Network **120** may be any network suitable for connecting operator device **110** with central communication system **130** and/or autonomous mobile robot **140**. Exemplary networks may include a local area network, a wide area network, the Internet, an ad hoc network, and so on. In some embodiments, network **120** may be a closed network that is not connected to the Internet (e.g., to heighten security and prevent external parties from interacting with central communication system **130** and/or autonomous mobile robot **140**). Such embodiments may be particularly advantageous where client device **110** is within the boundaries of environment **100**.
(17) Central communication system **130** acts as a central controller for a fleet of one or more robots including autonomous mobile robot **140**. Central communication system **130** receives information

from the fleet or the operator device **110** and uses that information to make decisions about activity to be performed by the fleet. Central communication system **130** may be installed on one device, or may be distributed across multiple devices. Central communication system **130** may be located within environment **100** or may be located outside of environment **100** (e.g., in a cloud implementation).

(18) Autonomous mobile robot **140** may be any robot configured to act autonomously with respect to a command. For example, an autonomous mobile robot **140** may be commanded to move an object from a source area to a destination area, and may be configured to make decisions autonomously as to how to optimally perform this function (e.g., which side to lift the object from, which path to take, and so on). Autonomous mobile robot **140** may be any robot suitable for performing a commanded function. Exemplary autonomous mobile robots include vehicles, such as forklifts, mobile storage containers, etc. References to autonomous mobile robot **140** in the singular are made for convenience and are non-limiting; these references equally apply to scenarios including multiple autonomous mobile robots.

(19) Exemplary Central Communication System Configuration

(20) FIG. **2** illustrates exemplary modules and data stores used by the central communication system **130**, in accordance with some embodiments. The central communication system **130** may include a source area module **231**, a destination area module **232**, a robot selection module **233**, and a robot instruction module **234**, as well as an environment map database **240**. The modules and databases depicted in FIG. **2** are merely exemplary; fewer or more modules and/or databases may be used by central communication system **130** to achieve the functionality disclosed herein. Additionally, the functionality of each component may be divided between the components differently from the description below. Additionally, each component may perform their respective functionalities in response to a request from a human, or automatically without human intervention.

(21) The source area module **231** identifies a source area. The term source area, as used herein, may refer to either a single point, several points, or a region surrounded by a boundary (sometimes referred to herein as a source boundary) within which a robot is to manipulate objects (e.g., pick up objects for transfer to another area). In an embodiment, the source area module **231** receives input from operator device **110** that defines the point(s) and/or region that form the source area. In an embodiment, the source area module **231** may receive input from one or more robots (e.g., image and/or depth sensor information showing objects known to need to be moved (e.g., within a predefined load dock)), and may automatically determine a source area to include a region within a boundary that surrounds the detected objects. The source area may change dynamically as objects are manipulated (e.g., the source area module **232** may shrink the size of the source area by moving boundaries inward as objects are transported out of the source area, and/or may increase the size of the source area by moving boundaries outward as new objects are detected).

(22) The destination area module **232** identifies a destination area. The term destination area, as used herein, may refer to either a single point, several points, or a region surrounded by a boundary (sometimes referred to herein as a destination boundary) within which a robot is to manipulate objects (e.g., drop an object off to rest). For example, where the objects are pallets in a warehouse setting, the destination area may include several pallet stands at different points in the facility, any of which may be used to drop off a pallet. The destination area module **232** may identify the destination area in any manner described above with respect to a source area, and may also identify the destination area using additional means.

(23) The destination area module **232** may determine the destination area based on information about the source area and/or the objects to be transported. Objects in the source area may have certain associated rules that add constraints to the destination area. For example, there may be a requirement that the objects be placed in a space having a predefined property (e.g., a pallet must be placed on a pallet stand, and thus the destination area must have a pallet stand for each pallet to be moved). As another example, there may be a requirement that the objects be placed at least a

threshold distance away from the destination area boundary, and thus, destination area module **232** may require a human draw the boundary at least at this distance and/or may populate the destination boundary automatically according to this rule (and thus, the boundary must be drawn at least that distance away). Yet further, destination area module **232** may require that the volume of the destination area is at least large enough to accommodate all of the objects to be transported that are initially within the source area.

(24) Source area module **231** and destination area module **232** may, in addition to, or alternative to, using rules to determine their respective boundaries, may use machine learning models to determine their respective boundaries. The models may be trained to take information as input, such as some or all of the above-mentioned constraints, sensory data, map data, object detection data, and so on, and to output boundaries based thereon. The models may be trained using data on tasks assigned to robots in the past, such as data on how operators have defined or refined the tasks based on various parameters and constraints.

(25) Robot selection module **233** selects one or more robots that are to transport objects from the source area to the destination area. In an embodiment, robot selection module **233** performs this selection based on one or more of a manipulation capability of the robots and a location of the robots within the facility. The term manipulation capability, as used herein, refers to a robot's ability to perform a task related to manipulation of an object. For example, if an object must be lifted, the robot must have the manipulation capability to lift objects, to lift an object having at least the weight of the given object to be lifted, and so on. Other manipulation capabilities may include an ability to push an object, an ability to drive an object (e.g., a mechanical arm may have an ability to lift an object, but may be unable to drive an object because it is affixed to, e.g., the ground), and so on. Further manipulation capabilities may include lifting and then transporting objects, hooking and then towing objects, tunneling and then transporting objects, using robots in combination with one another (e.g., an arm or other manipulates an object (e.g., lifts), places on another robot, and the robot then drives to the destination with the object). These examples are merely exemplary and non-exhaustive. Robot selection module **233** may determine required manipulation capabilities to manipulate the object(s) at issue, and may select one or more robots that satisfy those manipulation capabilities.

(26) In terms of location, robot selection module **233** may select one or more robots based on their location to the source area and/or the destination area. For example, robot selection module **233** may determine one or more robots that are closest to the source area, and may select those robot(s) to manipulate the object(s) in the source area. Robot selection module **233** may select the robot(s) based on additional factors, such as a number of objects to be manipulated, manipulation capabilities of the robot (e.g., how many objects can the robot carry at once; sensors the robot is equipped with; etc.), motion capabilities of the robot, and so on. In an embodiment, robot selection module **233** may select robots based on a state of one or more robot's battery (e.g., a closer robot may be passed up for a further robot because the closer robot has insufficient battery to complete the task). In an embodiment, robot selection module **233** may select robots based on their internal health status (e.g., where a robot is reporting an internal temperature close to overheating, that robot may be passed up even if it otherwise optimal, to allow that robot to cool down). Other internal health status parameters may include battery or fuel levels, maintenance status, and so on. Yet further factors may include future orders, a scheduling strategy that incorporates a longer horizon window (e.g., a robot that is optimal to be used now may, if used now, result in inefficiencies (e.g., depleted battery level or sub-optimal location), given a future task for that robot), a scheduling strategy that incorporates external processes, a scheduling strategy that results from information exchanged between higher level systems (e.g., WMS, ERP, EMS, etc.), and so on.

(27) The robot selection module **233** may select a robot using machine learning model trained to take various parameters as input, and to output one or more robots best suited to the task. The

inputs may include available robots, their manipulation capabilities, their locations, their state of health, their availability, task parameters, scheduling parameters, map information, and/or any other mentioned attributes of robots and/or tasks. The outputs may include an identification of one or more robots to be used (or suitable to be used) to execute a task. The robot selection module **233** may automatically select one or more of the identified robots for executing a task, or may prompt a user of operator device **110** to select from the identified one or more robots.

(28) The robot instruction module **234** transmits instructions to the selected one or more robots to manipulate the object(s) in the source area (e.g., to ultimately transport the object(s) to the destination area). In an embodiment, the robot instruction module **234** includes detailed step-by-step instructions on how to transport the objects. In another embodiment, the robot instruction module **234** transmits a general instruction to transmit one or more objects from the source area to the destination area, leaving the manner in which the objects will be manipulated and ultimately transmitted up to the robot to determine autonomously.

(29) The robot instruction module **234** may transmit instructions to a robot to traverse from a start pose to an end pose. In some embodiments, the robot instruction module **234** simply transmits a start pose and end pose to the robot and the robot determines a path from the start pose to the end pose. Alternatively, the robot instruction module **234** may provide some information on a path the robot should take to travel from the start pose to the end pose. Robot pathfinding is discussed in more detail below.

(30) Environment map database **240** stores information about the environment of the autonomous mobile robot **140**. The environment of an autonomous mobile robot **140** is the area within which the autonomous mobile robot **140** operates. For example, the environment may be a facility or a parking lot within which the autonomous mobile robot **140** operates. In some embodiments, the environment map database **240** stores environment information in one or more maps representative of the environment. The maps may be two-dimensional, three dimensional, or a combination of both. Central communication facility **130** may receive a map from operator device **110**, or may generate one based on input received from one or more robots **140** (e.g., by stitching together images and/or depth information received from the robots as they traverse the facility, and optionally stitching in semantic, instance, and/or other sensor-derived information into corresponding portions of the map). In some embodiments, the map stored by the environment map database **240** indicates the locations of obstacles within the environment. The map may include information about each obstacle, such as whether the obstacle is an animate or inanimate object.

(31) Environment map database **240** may be updated by central communication facility **130** based on information received from the operator device **110** or from the robots **140**. Information may include images, depth information, auxiliary information, semantic information, instance information, and any other information described herein. The environment information may include information about objects within the facility, obstacles within the facility, and auxiliary information describing activity in the facility. Auxiliary information may include traffic information (e.g., a rate at which humans and/or robots access a given path or area within the facility), information about the robots within the facility (e.g., manipulation capability, location, etc.), time-of-day information (e.g., traffic as it is expected during different segments of the day), and so on.

(32) The central communication facility **130** may continuously update environment information stored by the environment map database **240** as such information is received (e.g., to show a change in traffic patterns on a given path). The central communication facility **130** may also update environment information responsive to input received from the operator device **110** (e.g., manually inputting an indication of a change in traffic pattern, an area where humans and/or robots are prohibited, an indication of a new obstacle, and so on).

(33) Exemplary Autonomous Mobile Robot Configuration

(34) FIG. **3** illustrates exemplary modules and data stores used by the autonomous mobile robot, in

accordance with some embodiments. As depicted in FIG. **3**, autonomous mobile robot **140** includes a robot sensor module **331**, an object identification module **332**, a robot movement module **337**, and a robot navigation module **334**. The modules and databases depicted in FIG. **3** are merely exemplary; fewer or more modules and/or databases may be used to achieve the functionality described herein. Furthermore, any of the described functionality of the modules may instead be performed by the central communication system **130** or the operator device **110**. Additionally, any of the functionality of these modules may be performed with or without human instruction.

(35) The robot sensor module **331** includes a number of sensors that the robot uses to collect data about the robot's surroundings. For example, the robot sensor module **331** may include one or more cameras, one or more depth sensors, one or more scan sensors (e.g., RFID), a location sensor (e.g., showing location of the robot within the facility and/or GPS coordinates), and so on. Additionally, the robot sensor module **331** may include software elements for preprocessing sensor data for use by the robot. For example, the robot sensor module **331** may generate depth data information based on LIDAR sensor data. Data collected by the robot sensor module **331** may be used by the object identification module **332** to identify obstacles around the robot or may be used to determine a pose into which the robot must travel to reach an end pose.

(36) The object identification module **332** ingests information received from the robot sensor module **331**, and outputs information that identifies an object in proximity to the robot. The object identification module **332** may utilize information from a map of the facility (e.g., as retrieved from environment map database **240**) in addition to information from the robot sensor module **331** in identifying the object. For example, the object identification module **332** may utilize location information, semantic information, instance information, and so on to identify the object.

(37) Additionally, the object identification module **332** identifies obstacles around the robot for the robot to avoid. For example, the object identification module **332** determines whether an obstacle is an inanimate obstacle (e.g., a box, a plant, or a column) or an animate object (e.g., a person or an animal). The object identification module **332** may use information from the environment map database **240** to determine where obstacles are within the robot's environment. Similarly, the object identification module **332** may use information from the robot sensor module **331** to identify obstacles around the robot.

(38) The robot movement module **333** transports the robot within its environment. For example, the robot movement module **333** may include a motor, wheels, tracks, and/or legs for moving. The robot movement module **333** may include components that the robot uses to move from one pose to another pose. For example, the robot may use components in the robot movement module **333** to change its x-, y-, or z-coordinates or to change its orientation. In some embodiments, the robot movement module **333** receives instructions from the robot navigation module **334** to follow a path determined by the robot navigation module **334** and performs the necessary actions to transport the robot along the determined path.

(39) The robot navigation module **334** determines a path for the robot from a start pose to an end pose within the environment. A pose of the robot may refer to an orientation of the robot and/or a location of the robot (including x-, y-, and z-coordinates). The start pose may be the robot's current pose or some other pose within the environment. The end pose may be an ultimate pose within the environment to which the robot is traveling or may be an intermediate pose between the ultimate pose and the start pose. The path may include a series of instructions for the robot to perform to reach the goal pose. For example, the path may include instructions for the robot to travel from one x-, y-, or z-coordinate to another and/or to adjust the robot's orientation (e.g., by taking a turn or by rotating in place). In some embodiments, the robot navigation module **334** implements routing instructions received by the robot from the central communication system **130**. For example, the central communication system **130** may transmit an end pose to the robot navigation module **334** or a general path for the robot to take to a goal pose, and the robot navigation module **334** may determine a path that avoids objects, obstacles, or people within the environment. The robot

navigation module **334** may determine a path for the robot based on sensor data or based on environment data. In some embodiments, the robot navigation module **334** updates an already determined path based on new data received by the robot.

(40) In some embodiments, the robot navigation module **334** receives an end location and the robot navigation module **334** determines an orientation of the robot necessary to perform a task at the end location. For example, the robot may receive an end location and an instruction to deliver an object at the end location, and the robot navigation module **334** may determine an orientation that the robot must take to properly deliver the object at the end location. The robot navigation module **334** may determine a necessary orientation at the end location based on information captured by the robot sensor module **331**, information stored by the environment map database **240**, or based on instructions received from an operator device. In some embodiments, the robot navigation module **334** uses the end location and the determined orientation at the end location to determine the end pose for the robot.

Example Automated Trailer Loading and Unloading

(41) FIG. **4** illustrates an example autonomous mobile robot **400** navigating through a warehouse environment to a trailer **410**, in accordance with some embodiments. The autonomous mobile robot **400** may have been instructed by a central communication system to load or unload the trailer **410**. The central communication system also may have specified an area **420** to which autonomous mobile robot **400** may unload objects from the trailer **410** or from which the autonomous mobile robot **400** may obtain objects to load into the trailer **410**. The central communication system may also specify which of a set docking points **430** the trailer **410** has docked to.

(42) When the autonomous mobile robot reaches the docking point, the autonomous mobile robot may use sensor data to collect information about the trailer. The sensor data is data collected by one or more sensors on the autonomous mobile robot. For example, the sensor data may include LIDAR data captured by a LIDAR sensor and/or image data captured by a camera. In some embodiments, the sensor data includes data that describes the locations of objects and obstacles around the autonomous mobile robot in a two-dimensional space and/or a three-dimensional space. The autonomous mobile robot may determine characteristics of the trailer based on the sensor data. For example, the autonomous mobile robot may determine the width, height, depth, centerline, off-center parking, yaw, roll, and/or pitch of the trailer.

(43) Additionally, the autonomous mobile robot may determine its location relative to the trailer based on the sensor data. For example, the autonomous mobile robot may identify some point or part of the trailer and determine its location with respect to that point or part. The autonomous mobile robot may use the sensor data determine its location and orientation relative the trailer. Furthermore, the autonomous mobile robot may identify objects in the trailer and may determine poses of the objects. For example, the autonomous mobile robot may identify pallets, including the types of the pallets, and may determine their location and orientation within the trailer. In some embodiments, the autonomous mobile robot continually determines its location, and the locations of objects and obstacles, based on continually received sensor data. The autonomous mobile robot may continually receive sensor data on a regular or irregular basis.

(44) In some embodiments, the autonomous mobile robot uses a machine-learning model (e.g., a neural network) to determine characteristics of the trailer based on sensor data. For example, the machine-learning model may be a computer-vision model that has been trained to determine characteristics of a trailer based on image data captured by a camera on the autonomous mobile robot. Similarly, the machine-learning model may be trained to determine the location and orientation of objects within the trailer based on sensor data.

(45) The autonomous mobile robot may enter the trailer and use sensor data of the trailer to determine the autonomous mobile robot's location with respect to the trailer. Additionally, the autonomous mobile robot may determine the location of the objects in the trailer, and any obstacles in the trailer, with respect to the trailer based on the sensor data. The autonomous mobile robot

identifies an object to unload from the trailer and manipulates the object using a forklift component. In some embodiments, while the autonomous mobile robot navigates within the trailer, the autonomous mobile robot travels slightly offset from a centerline of the trailer so that the autonomous mobile robot is more likely to be in a correct position to manipulate an object within the trailer. In these embodiments, by remaining slightly offset from the centerline of the trailer, the autonomous mobile robot will likely be able to position its forks to lift an object by simply side-shifting its forks.

(46) In some embodiments, the autonomous mobile robot uses an enhanced navigation algorithm while navigating within the trailer. An enhanced navigation algorithm may enable the autonomous mobile robot to determine its location more accurately within the trailer and to determine more accurately the locations of objects and obstacles within the trailer. The enhanced navigation algorithm may be more precise than a navigation algorithm used by the autonomous mobile robot while the autonomous mobile robot navigates within the warehouse environment. In some embodiments, the enhanced navigation algorithm uses a map within the trailer that has a finer resolution than the environment map. Similarly, the enhanced navigation algorithm may use denser motion primitives to navigate within the trailer than a navigation algorithm used by the autonomous mobile robot when navigating within the warehouse environment. In some embodiments, the enhanced navigation algorithm uses a modified version of A* search to navigate within the trailer. Furthermore, the enhanced navigation algorithm may use sensor data with a narrower field of view than the sensor data used a navigation algorithm used while the autonomous mobile robot navigates within the warehouse environment.

(47) The autonomous mobile robot may detect when it has entered the trailer and may start using an enhanced navigation algorithm upon determining that it has entered the trailer. The autonomous mobile robot may use the enhanced navigation algorithm to position itself to manipulate objects within the trailer and to transport an object out of the trailer. Furthermore, the autonomous mobile robot may detect when it has exited the trailer and stop using an enhanced navigation algorithm upon determining that it is no longer in the trailer.

(48) In some embodiments, the autonomous mobile robot uses a first navigation algorithm to determine a route from a first pose in the warehouse environment to a second pose near an entrance to the trailer. The autonomous mobile robot may then use a second navigation algorithm to determine a route from the second pose to a third pose within the trailer from which the autonomous mobile robot can manipulate an object. The first navigation algorithm may be a navigation algorithm that the autonomous mobile robot uses to navigate within the warehouse environment and the second navigation algorithm may be an enhanced navigation algorithm that the autonomous mobile robot uses to determine its location within the trailer. Thus, the second navigation algorithm may be a navigation algorithm with a higher level of precision than the first navigation algorithm. For example, the second navigation algorithm may use a map with a finer resolution or may use more precise or dense motion primitives to determine a route.

(49) FIGS. **5**A and **5**B illustrate an example autonomous mobile robot **500** navigating within a trailer **510** and determining characteristics of the trailer **510**, in accordance with some embodiments. FIG. **5**A illustrates example characteristics that the autonomous mobile robot **500** may determine based on sensor data. For example, the autonomous mobile robot **500** may determine a width **520** and a depth **530** of the trailer **510**. Additionally, the autonomous mobile robot **20** may determine the location of a centerline **540** of the trailer **510**.

(50) FIG. **5**B illustrates example locations **560** of the autonomous mobile robot **500** and objects **550** within the trailer **510**, as determined by the autonomous mobile robot **500**. The autonomous mobile robot **500** determines the locations **560** of itself and objects **550** relative to the trailer **510**. In the embodiment illustrated in FIG. **5**B, the locations **560** of the autonomous mobile robot **500** and the objects **550** are determined relative to an origin location **570**, however alternative embodiments may determine locations **560** within the trailer **510** relative to any part of the trailer

510.

(51) The autonomous mobile robot generally must unload the last row of the trailer first. In some embodiments, the autonomous mobile robot determines whether the trailer is full and, responsive to determining that the trailer is full, determines that it must unload the last row of the trailer. As used herein, the last row of the trailer is the row that is closest to the doors through which the autonomous mobile robot enters to load or unload the trailer. When the autonomous mobile robot prepares to unload the last row of the trailer, the autonomous mobile robot generally must manipulate the objects in the last row from a ramp that connects the warehouse to the trailer. Therefore, the autonomous mobile robot accounts for the angle of the ramp while approaching the objects.

(52) The autonomous mobile robot uses sensor data to monitor its location relative to the object. Additionally, the autonomous mobile robot may use accelerometer or gyroscopic data to determine its orientation on the ramp and to thereby adjust the orientation of a forklift component coupled to the autonomous mobile robot. The autonomous mobile robot adjusts its forklift components to ensure that the forks are level with respect to the floor of the trailer, rather than the ramp. The autonomous mobile robot thereby ensures that the forks are in the correct orientation to manipulate a pallet. The robot may continually adjust its forks as it approaches the object. For example, the autonomous mobile robot may continually gather new sensor data about its surroundings to continually determine its location and pose with respect to an object in the last row of the trailer and continually adjust its forklift component.

(53) FIGS. **6**A-**6**C illustrate an example autonomous mobile robot **600** unloading objects in the last row of a trailer **610**, in accordance with some embodiments. FIG. **6**A illustrates an autonomous mobile robot **600** approaching a ramp **620** that leads from the warehouse to the trailer **610**. FIG. **6**B illustrates an autonomous mobile robot **600** attempting to unload an object in the last row of the trailer **610** without adjusting its forklift **630**. The forklift **630** remains level with respect to the ramp **620**, meaning that the forks of the forklift **630** would collide **640** with the floor of the trailer **610**. FIG. **6**C illustrates an autonomous mobile robot **600** that adjusts its forklift **630** to be level with the floor of the trailer **610**. By continually adjusting its forklift **630** as it travels down the ramp **620**, the autonomous mobile robot **600** ensures that its forklift **630** is able to manipulate an object in the last row of the trailer **610** without having to maneuver itself significantly.

(54) While performing an unload mission, the autonomous mobile robot continues to unload objects from a trailer to a destination area within the warehouse until the trailer is empty. In some embodiments, the autonomous mobile robot determines that the trailer is empty by navigating to the trailer, collecting sensor data of the interior of the trailer, and determining that there are no more objects in the trailer.

(55) In some embodiments, the autonomous mobile robot continues to determine its location with respect to environment map of the warehouse while also determining its location with respect to the trailer. The autonomous mobile robot may use its location with respect to the trailer for navigating within the trailer. However, by continuing to track its location with respect to the environment map, the autonomous mobile robot can more quickly return to navigating with respect to the environment map.

(56) To continue to determine its location with respect to the environment map while navigating within the trailer, the autonomous mobile robot may determine the location of the trailer with respect to the warehouse. For example, the autonomous mobile robot may capture images of the entrance of the trailer and/or of the entire exterior of the trailer. The autonomous mobile robot may then determine the location of the trailer with respect to the warehouse based on these captured images. The autonomous mobile robot may then continue to determine its location with respect to the environment map based on the determined location of the trailer with respect to the warehouse. The autonomous mobile robot performs a mission to load a trailer in a similar manner to how the autonomous mobile robot unloads a trailer. In some embodiments, the autonomous mobile robot

performs an "empty run" of the trailer, where the autonomous mobile robot travels into a trailer that is to be loaded to collect sensor data and determine characteristics of the trailer. The autonomous mobile robot may perform this "empty run" without carrying any objects from a source area.
(57) To load the trailer, the autonomous mobile robot identifies an object in a source area to load onto the trailer. The autonomous mobile robot picks up the object and navigates from the source area to the trailer. The autonomous mobile robot determines whether the object needs to be loaded in the last row of the trailer. If the object does not need to be placed in the last row of the trailer, the autonomous mobile robot enters the trailer and identifies a location in the trailer where the object will be placed. In some embodiments, the autonomous mobile robot travels within the trailer slightly offset from a centerline of the trailer so that the autonomous mobile robot is more likely to be in a correct pose to deliver the object to a proper location within the trailer.
(58) If the object needs to be placed in the last row of the trailer, the autonomous mobile robot will continually collect sensor data to determine a correct orientation of its forklift such that the object is delivered level with the floor of the trailer. The autonomous mobile robot continually adjusts the orientation of its forklift until the autonomous mobile robot delivers the object to a proper spot in the last row of the trailer.
(59) The autonomous mobile robot may include components that enable the autonomous mobile robot to navigate within the trailer. For example, the autonomous mobile robot may include a movement system that enables the autonomous mobile robot to rotate in place. Similarly, the autonomous mobile robot may be configured to move to perpendicularly to the direction it is facing without changing its orientation. Thus, the autonomous mobile robot is capable of navigating within often narrow spaces within a trailer and to position itself to be able to manipulate objects within the trailer.
(60) Furthermore, the autonomous mobile robot may include components that enable the autonomous mobile robot to manipulate objects within the trailer. For example, the autonomous mobile robot may include a forklift with the ability to side-shift. Thus, the autonomous mobile robot can position the forklift to manipulate objects without having to reposition itself. The autonomous mobile robot may also include an arm that can lift objects while the autonomous mobile robot maintains a fixed pose. Thus, the autonomous mobile robot can effectively manipulate objects within the trailer from the often limited poses available within the trailer.
(61) Example Centerline Heuristic-Based Path Finding
(62) FIG. **7** illustrates an example autonomous mobile robot that uses a centerline heuristic to navigate within a trailer, in accordance with some embodiments. While the FIG. **7** illustrates an autonomous mobile robot entering a trailer, a similar process may be used for an autonomous mobile robot exiting the trailer, as well.
(63) The autonomous mobile robot **700** receives sensor data describing the environment around the robot. The autonomous mobile robot **700** may receive the sensor data from sensors coupled to the robot (e.g., the robot sensor module **331**) or from sensors remote from the autonomous mobile robot.
(64) The autonomous mobile robot **700** determines an initial pose **710** of the autonomous mobile robot **700** and an end pose **720** for the autonomous mobile robot **700** based on the received sensor data. The initial pose **710** of the autonomous mobile robot **700** may be a current pose of the autonomous mobile robot, or a pose near the entrance of a trailer **730**. The end pose **720** may be a pose for handling an object **740** (e.g., an item on a pallet) within the trailer **730**. In some embodiments, the autonomous mobile robot **700** determines the end pose based on instructions from a central communications system.
(65) The autonomous mobile robot **700** computes a centerline **750** of the trailer **730** based on the received sensor data. The centerline is a centerline of the trailer (i.e., is a line that is equidistant from the sides of the trailer). In some embodiments, the autonomous mobile robot **700** generates a map of the interior of the trailer based on the sensor data and computes the centerline based on the

generated map.

(66) FIG. **7** illustrates an example path **760** from the initial pose **710** to the end pose **720** that is computed without a centerline heuristic. The autonomous mobile robot may compute the path **760** using a pathfinding algorithm, such as Hybrid A* or State Lattice Planner. A pathfinding algorithm may use heuristics to compute a path from a first pose to a second pose. For example, a pathfinding algorithm may use a heuristic to represent a cost of a set of nodes through which the autonomous mobile robot **700** may pass as part of the path. These nodes may be coordinates or locations within the trailer through which the robot may pass as part of a computed path. The heuristic may have a higher cost for a “worse” node than for a “better” node. For example, a common heuristic is the distance from a node to the end pose, and the heuristic may thereby compute a higher cost (i.e., distance) for a node that is further away from the end pose than a node that is closer to the end pose. The pathfinding algorithms compute a path through nodes that minimize the total cost of the nodes that the path passes through.

(67) As illustrated in FIG. **7**, a path **760** that uses traditional heuristics is likely to take a more direct path to the end pose **720**. While this may mean that the autonomous mobile robot **700** takes a shorter path, the path **760** may take the autonomous mobile robot **700** closer to the sides of the trailer **730**, especially if the end pose **720** is closer to one side of the trailer **730** than the other. This may reduce the space that the autonomous mobile robot **700** has to maneuver if the autonomous mobile robot **700** has to change course within the trailer **730**.

(68) FIG. **7** illustrates a path **770** computed using a centerline heuristic. A centerline heuristic is a heuristic that represents a distance of nodes to the centerline **750**. For example, the centerline heuristic may increase the cost of a node that is further from the centerline **750** relative to a node that is closer to the centerline **750**. In some embodiments, the centerline heuristic assigns a cost value to a node that is proportional to the distance of the node to the centerline of the trailer. Alternatively, the centerline heuristic may assign a constant cost value to nodes within some threshold distance of the centerline, and may assign cost values to other nodes that is proportion to their distance from the centerline minus the threshold distance. In other words, the centerline heuristic may assign cost values to nodes such that nodes within some area (e.g., a rectangular area) around the centerline are constant, and that nodes outside the area are assigned cost values that are proportional to their distance from the area.

(69) The autonomous mobile robot **700** may use a combination of multiple heuristics to compute the path **770** to the end pose **720**. For example, the autonomous mobile robot **700** may compute a combination of the centerline heuristic, a heuristic representing a distance of a node to the end pose **720**, and a heuristic representing a distance of a node to an obstacle. Thus, the autonomous mobile robot **700** may compute a combined cost for each node that is a function (e.g., a linear combination) of multiple heuristics.

(70) As explained above, the autonomous mobile robot **700** computes the path **770** from the initial pose **710** to the end pose **720** based on the centerline heuristic. The autonomous mobile robot **700** travels along the path **770** to the end pose **720**. The autonomous mobile robot **700** may manipulate an object **740** at the end pose **720**, such as lifting the object to transport. The autonomous mobile robot **700** may perform a similar process to compute a path to exit the trailer **730**.

(71) Example Method for Automated Trailer Loading and Unloading

(72) FIG. **8** is a flowchart for an example method for automated trailer loading and unloading, in accordance with some embodiments. Alternative embodiments may include more, fewer, or different steps, and the steps may be performed in a different order from that illustrated in FIG. **8**. Additionally, while FIG. **8** is primarily described in terms of actions taken by an autonomous mobile robot, the steps may be performed, in part or in whole, by a central communication system in communication with an autonomous mobile robot.

(73) The autonomous mobile robot determines **800** a location of a trailer with respect to an environment (e.g., a warehouse). The autonomous mobile robot is caused **810** to travel from a first

pose (e.g., a current pose of the autonomous mobile robot) to a second pose near an entrance of the trailer. This second pose may be on or near a loading ramp that bridges the environment and the trailer.

(74) The autonomous mobile robot receives **820** sensor data from a sensor while at the second pose. The sensor data describes the trailer and an object within a cargo area of the trailer. The autonomous mobile robot determines **830** a location of the autonomous mobile robot with respect to the trailer based on the sensor data. The autonomous mobile robot also determines **840** a location of the object with respect to the trailer based on the sensor data. The autonomous mobile robot is caused **850** to travel from the second pose to a third pose within the trailer based on the sensor data, the determined location of the autonomous mobile, and the determined location of the object. The third pose is a pose from which the autonomous mobile robot can manipulate the object.

(75) Example Method for Pathfinding Using a Centerline Heuristic

(76) FIG. **9** is a flowchart for an example method for pathfinding using a centerline heuristic, in accordance with some embodiments. Alternative embodiments may include more, fewer, or different steps, and the steps may be performed in a different order from that illustrated in FIG. **9**. Additionally, while FIG. **9** is primarily described in terms of actions taken by an autonomous mobile robot, the steps may be performed, in part or in whole, by a central communication system in communication with an autonomous mobile robot.

(77) An autonomous mobile robot receives **900** sensor data describing an environment (e.g., a warehouse) around the autonomous mobile robot. The autonomous mobile robot determines **910** an initial pose of the autonomous mobile robot relative to the trailer based on the sensor data and determines **920** an end pose for the autonomous mobile robot in the trailer based on the sensor data. The autonomous mobile robot computes **930** a centerline of the trailer based on the sensor data and computes **940** a path to the end pose. The autonomous mobile robot computes the centerline based on the sensor data and a centerline heuristic, where the centerline heuristic is a heuristic representing a distance of portions of the path from the centerline. The autonomous mobile robot is caused **950** to traverse the computed path to the end pose.

(78) Additional Considerations

(79) The foregoing description of the embodiments has been presented for the purpose of illustration; many modifications and variations are possible while remaining within the principles and teachings of the above description.

(80) Any of the steps, operations, or processes described herein may be performed or implemented with one or more hardware or software modules, alone or in combination with other devices. In some embodiments, a software module is implemented with a computer program product comprising one or more computer-readable media storing computer program code or instructions, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described. In some embodiments, a computer-readable medium comprises one or more computer-readable media that, individually or together, comprise instructions that, when executed by one or more processors, cause the one or more processors to perform, individually or together, the steps of the instructions stored on the one or more computer-readable media. Similarly, a processor comprises one or more processors or processing units that, individually or together, perform the steps of instructions stored on a computer-readable medium.

(81) Embodiments may also relate to a product that is produced by a computing process described herein. Such a product may store information resulting from a computing process, where the information is stored on a non-transitory, tangible computer-readable medium and may include any embodiment of a computer program product or other data combination described herein.

(82) The description herein may describe processes and systems that use machine learning models in the performance of their described functionalities. A “machine learning model,” as used herein, comprises one or more machine learning models that perform the described functionality. Machine learning models may be stored on one or more computer-readable media with a set of weights.

These weights are parameters used by the machine learning model to transform input data received by the model into output data. The weights may be generated through a training process, whereby the machine learning model is trained based on a set of training examples and labels associated with the training examples. The training process may include: applying the machine learning model to a training example, comparing an output of the machine learning model to the label associated with the training example, and updating weights associated for the machine learning model through a back-propagation process. The weights may be stored on one or more computer-readable media, and are used by a system when applying the machine learning model to new data.
(83) The language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to narrow the inventive subject matter. It is therefore intended that the scope of the patent rights be limited not by this detailed description, but rather by any claims that issue on an application based hereon.
(84) As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive “or” and not to an exclusive “or”. For example, a condition “A or B” is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present). Similarly, a condition “A, B, or C” is satisfied by any combination of A, B, and C being true (or present). As a not-limiting example, the condition “A, B, or C” is satisfied when A and B are true (or present) and C is false (or not present). Similarly, as another not-limiting example, the condition “A, B, or C” is satisfied when A is true (or present) and B and C are false (or not present).

Claims

1. A method comprising: determining a location of a trailer with respect to an environment; causing an autonomous mobile robot within the environment to travel from a first pose to a second pose within a threshold distance of an entrance to the trailer, wherein the second pose is within the threshold distance of a loading ramp that bridges the environment and the trailer, and wherein causing the autonomous mobile robot to travel from the first pose to the second pose comprises: determining a first route from the first pose to the second pose by applying a first navigation algorithm that uses a first set of motion primitives for the autonomous mobile robot; receiving sensor data from a sensor of the autonomous mobile robot at the second pose, the sensor data describing the trailer and an object loaded within a cargo area of the trailer; determining, based on the sensor data, a location of the autonomous mobile robot with respect to the trailer; determining, based on the sensor data, a location of the object with respect to the trailer; causing the autonomous mobile robot to travel from the second pose to a third pose within the trailer based on the sensor data, the location of the autonomous mobile robot with respect to the trailer, and the location of the object with respect to the trailer, wherein the third pose is a pose from which the autonomous mobile robot can manipulate the object, and wherein causing the autonomous mobile robot to travel from the second pose to the third pose comprises: determining a second route from the second pose to the third pose by applying a second navigation algorithm, wherein the second navigation algorithm uses a second set of motion primitives for the autonomous mobile robot, and wherein the second navigation algorithm achieves a higher precision relative to the first navigation algorithm because the second set of motion primitives includes more motion primitives than the first set of motion primitives.
2. The method of claim 1, wherein the second navigation algorithm achieves a higher precision than the first navigation algorithm by using a map with a finer resolution than a map used by the

first navigation algorithm.
3. The method of claim 1, wherein one of the first navigation algorithm or the second navigation algorithm includes an A* search algorithm.
4. The method of claim 1, further comprising: continually receiving second sensor data from a second sensor of the autonomous mobile robot while the autonomous mobile robot travels to the second pose; determining, based on the continually received second sensor data, an orientation of a floor of the trailer with respect to the autonomous mobile robot; and responsive to continually receiving second sensor data, continually adjusting a forklift of the autonomous mobile robot based on the continually received sensor data while the autonomous mobile robot travels to the second pose.
5. The method of claim 1, wherein the environment comprises a warehouse.
6. A non-transitory computer-readable medium storing instructions that, when executed by a processor, cause the processor to: determine a location of a trailer with respect to an environment; cause an autonomous mobile robot within the environment to travel from a first pose to a second pose within a threshold distance of an entrance to the trailer, wherein the second pose is within the threshold distance of a loading ramp that bridges the environment and the trailer, and wherein causing the autonomous mobile robot to travel from the first pose to the second pose comprises: determining a first route from the first pose to the second pose by applying a first navigation algorithm that uses a first set of motion primitives for the autonomous mobile robot; receive sensor data from a sensor of the autonomous mobile robot at the second pose, the sensor data describing the trailer and an object loaded within a cargo area of the trailer; determine, based on the sensor data, a location of the autonomous mobile robot with respect to the trailer; determine, based on the sensor data, a location of the object with respect to the trailer; cause the autonomous mobile robot to travel from the second pose to a third pose within the trailer based on the sensor data, the location of the autonomous mobile robot with respect to the trailer, and the location of the object with respect to the trailer, wherein the third pose is a pose from which the autonomous mobile robot can manipulate the object, and wherein causing the autonomous mobile robot to travel from the second pose to the third pose comprises: determining a second route from the second pose to the third pose by applying a second navigation algorithm, wherein the second navigation algorithm uses a second set of motion primitives for the autonomous mobile robot, and wherein the second navigation algorithm achieves a higher precision relative to the first navigation algorithm because the second set of motion primitives includes more motion primitives than the first set of motion primitives.
7. The computer-readable medium of claim 6, wherein the second navigation algorithm achieves a higher precision than the first navigation algorithm by using a map with a finer resolution than a map used by the first navigation algorithm.
8. The computer-readable medium of claim 6, wherein one of the first navigation algorithm or the second navigation algorithm includes an A* search algorithm.
9. The computer-readable medium of claim 6, further storing instructions that cause the processor to: continually receive second sensor data from a second sensor of the autonomous mobile robot while the autonomous mobile robot travels to the second pose; determine, based on the continually received second sensor data, an orientation of a floor of the trailer with respect to the autonomous mobile robot; and responsive to continually receiving second sensor data, continually adjust a forklift of the autonomous mobile robot based on the continually received sensor data while the autonomous mobile robot travels to the second pose.
10. The computer-readable medium of claim 6, wherein the environment comprises a warehouse.
11. An autonomous mobile robot comprising a processor and a non-transitory computer-readable medium storing instructions that, when executed by the processor, cause the processor to: determine a location of a trailer with respect to an environment; cause the autonomous mobile robot within the environment to travel from a first pose to a second pose within a threshold distance

of an entrance to the trailer, wherein the second pose is within the threshold distance of a loading ramp that bridges the environment and the trailer, and wherein causing the autonomous mobile robot to travel from the first pose to the second pose comprises: determining a first route from the first pose to the second pose by applying a first navigation algorithm that uses a first set of motion primitives for the autonomous mobile robot; receive sensor data from a sensor of the autonomous mobile robot at the second pose, the sensor data describing the trailer and an object loaded within a cargo area of the trailer; determine, based on the sensor data, a location of the autonomous mobile robot with respect to the trailer; determine, based on the sensor data, a location of the object with respect to the trailer; cause the autonomous mobile robot to travel from the second pose to a third pose within the trailer based on the sensor data, the location of the autonomous mobile robot with respect to the trailer, and the location of the object with respect to the trailer, wherein the third pose is a pose from which the autonomous mobile robot can manipulate the object, and wherein causing the autonomous mobile robot to travel from the second pose to the third pose comprises: determining a second route from the second pose to the third pose by applying a second navigation algorithm, wherein the second navigation algorithm uses a second set of motion primitives for the autonomous mobile robot, and wherein the second navigation algorithm achieves a higher precision relative to the first navigation algorithm because the second set of motion primitives includes more motion primitives than the first set of motion primitives.

12. The autonomous mobile robot of claim 11, wherein the second navigation algorithm achieves a higher precision than the first navigation algorithm by using a map with a finer resolution than a map used by the first navigation algorithm.

13. The autonomous mobile robot of claim 11, wherein one of the first navigation algorithm or the second navigation algorithm includes an A* search algorithm.

14. The autonomous mobile robot of claim 11, wherein the computer-readable medium further stores instructions that cause the processor to: continually receive second sensor data from a second sensor of the autonomous mobile robot while the autonomous mobile robot travels to the second pose; determine, based on the continually received second sensor data, an orientation of a floor of the trailer with respect to the autonomous mobile robot; and responsive to continually receiving second sensor data, continually adjust a forklift of the autonomous mobile robot based on the continually received sensor data while the autonomous mobile robot travels to the second pose.